

3. (a) The table shows information about some organic compounds.

Name	Molecular formula	Structure	Homologous series
ethanol	C_2H_5OH		alcohols
ethanoic acid	CH_3COOH	$ \begin{array}{c} H \\ \\ H-C-C \begin{array}{l} \nearrow O \\ \searrow O-H \end{array} \\ \\ H \end{array} $	
.....	C_3H_6	$ \begin{array}{c} & H & & H \\ & & & \\ H & -C & =C & -C- \\ & & & \\ H & & & H \end{array} $	alkenes

(i) **Complete the table.** [3]

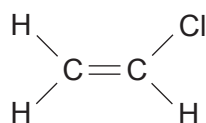
(ii) The molecular mass of an alkene is 98. Give the molecular formula of this alkene. [1]

$$A_r(C) = 12 \qquad A_r(H) = 1$$

Molecular formula

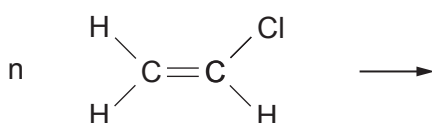


- (b) Polyvinyl chloride (PVC) is formed from the following monomer in a polymerisation reaction.



Complete the equation for the polymerisation reaction.

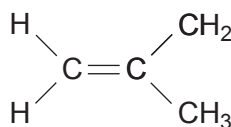
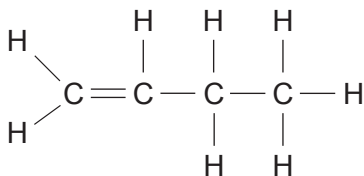
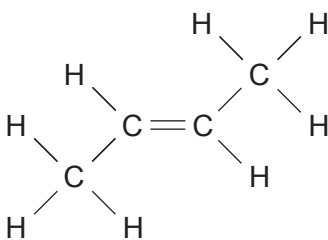
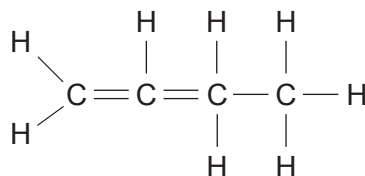
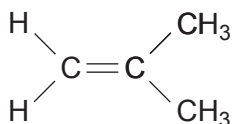
[2]



- (c) A student was asked to draw structures for isomers with the molecular formula C_4H_8 .

She drew the following diagrams. Circle **all** the **correct** structures.

[2]



(d) Pent-2-ene reacts with bromine water in a similar manner to ethene.

Complete the equation for the reaction of pent-2-ene with bromine by drawing the structure of the product.

[1]

